

INTERNAL · CONFIDENTIAL

Santander Business Banking

Concept Prototype — Technical Report

A front-end exploration of paperless, compliant-by-design service journeys

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SECTION 1

1 Executive summary

Business banking retains a striking dependence on paper. Mandate variations, account closures, beneficiary onboarding and dispute handling still routinely require posted forms, wet-ink signatures and branch attendance — processes that impose days of latency, measurable operational cost, and inconsistent application of controls at exactly the points where regulatory rigour matters most.

This report describes a self-initiated concept prototype that reframes seventeen such service events as guided, digitally-signed, self-serve journeys. The distinguishing idea is that the controls each journey must satisfy — Strong Customer Authentication [3], Confirmation of Payee [4], sanctions and politically-exposed-person screening [8], cooling-off, and record-keeping [12] — are expressed inside the flow rather than appended through manual back-office process. Control thereby becomes an enabler of a better experience instead of the reason to retain paper.

The prototype is deliberately a front-end artefact. It holds no customer data, connects to no live system, and is not a release candidate. Its purpose is to make a service proposition tangible enough to test — with colleagues, with risk and compliance, and with business customers — before any commitment to build.

Dimension	Summary
Workflows	Seventeen end-to-end paperless journeys
Entity coverage	Seven business types (sole trader, partnership, limited company, LLP, charity, club, society)
Platform	Single-file React front end, compiled to one self-contained HTML bundle
Regulatory framing	PSD2/SCA, CoP, PSR 2017, MLR 2017, FCA DISP/BCOBS/SYSC, Consumer Duty, UK GDPR, MTD
Scope	Front-end concept only — no backend, no real data, not production-hardened

SECTION 2

2 Context and problem statement

The friction in business-banking servicing is not primarily a technology gap; it is a process gap. The underlying capabilities — digital identity, strong authentication, Confirmation of Payee and real-time payments — are mature and, in several cases, mandated. Yet the service events that sit on top of them are frequently still discharged on paper, for three connected reasons: the controls are treated as separate from the customer journey; the journeys are owned by different teams; and no single artefact makes the end-to-end experience concrete enough to challenge.

The cost of that gap is cumulative. Postal round-trips convert minutes of decision-making into days of elapsed time. Manual re-keying introduces error and rework. And discretionary, manually-applied controls are, by construction, harder to evidence consistently — a growing concern under the FCA Consumer Duty [1], which expects firms to demonstrate good outcomes rather than assert them.

THE CORE QUESTION

What would these service events look like if they were re-imagined as in-app, self-serve, compliant-by-design journeys — with each regulatory check built into the flow, applied consistently, and recorded automatically?

SECTION 3

3 Objectives, scope and non-goals

The prototype was built to answer that question in a form that can be reviewed and tested, not merely described. Its objectives are therefore evidential rather than technical:

- Demonstrate credible paperless equivalents for the most paper-heavy business-banking service events.
- Express, within each journey, the specific regulatory control it must satisfy — so reviewers can see where an obligation is met rather than infer it.
- Provide a single, clickable artefact for internal review and structured customer research.
- Establish a consistent design language and a measurable accessibility standard for the journeys.

Non-goals (explicitly out of scope)

Equally important is what the prototype is not, so that its evidence is read correctly:

- No backend, database, or integration with live banking systems.
- No real customer, account, or payment data — every figure and reference is illustrative and fictional.
- No production security hardening; this is a concept artefact, not a release candidate.
- No processing of real payments, identity documents, or regulatory submissions.

SECTION 4

4 System architecture

The prototype is implemented as a single React component compiled by Vite into one self-contained HTML bundle. This is an unusual choice for production software but a deliberate one for a concept artefact: it keeps the entire product readable top-to-bottom, removes build-and-deploy friction, and lets the whole experience be shared as a single file. The trade-off — that all state lives in one component — is acceptable precisely because the artefact is not intended to scale as a codebase.

Architectural choice	Rationale
Single-file React front end	Whole product legible in one place; no module navigation to review it
Compiled to one HTML artefact	Frictionless to deploy, host, or hand to a colleague
Two shells, shared state (mobile & desktop)	Demonstrates responsive parity without a second codebase
Workflow-overlay model	Each journey is a self-contained, multi-step wizard over the main app
Entity system (7 business types)	Accounts, mandate rules, labels and required documents reconfigure per type

Mandate and authorisation model

Authorisation is modelled explicitly. Each account carries a signing rule — any-one, any-two, or all signatories — and every workflow resolves the strictest applicable rule across the accounts it touches. That resolved rule determines whether an instruction can proceed immediately, requires a co-signatory, or enters a cooling-off period. Encoding authorisation as data rather than as bespoke per-journey logic is what allows the same seventeen journeys to behave correctly across all seven entity types.

SECTION 5

5 Workflow catalogue

The prototype implements seventeen end-to-end workflows. Each replaces an identifiable paper or branch-based process and surfaces the control that governs it.

Workflow	What it replaces / does	Primary control
Account closure	Posted closure request; in-credit and cooling-off checks	BCOBS [11]
Business details update	Change of name/address/contact with proof of address	KYB [8]
Mandate changes	Paper mandate variation; add/remove signatories, signing rule	SM&CR; [12]
Bulk payments / wages	BACS paper schedule; payee book with per-payee CoP	CoP [4]
Pre-approved lending	Facility request by phone/post; offer and draw-down	CCA 1974 [13]
International FX payment	Faxed instruction; rate and fee disclosure	FTR [9]
Dormant reactivation	Branch reactivation of a dormant account	Dormancy handling
Personal / business unlink	Manual separation of personal data from a relationship	UK GDPR [15]
Credit ring-fence	Excludes personal data from business credit decisioning	UK GDPR Art. 5(1)(c) [15]
Signatory ID register	Paper KYC/KYB across identity and address evidence	MLR 2017 [8]; ICO [17]
MTD VAT submission	Manual VAT return; categorise, review, submit	HMRC MTD [14]
Complaint handling	Paper complaint; intake, triage, decision, FOS route	FCA DISP [10]
Standing orders & DDs	Paper mandate / cancellation letter	Direct Debit Guarantee [5]
Transaction dispute	Paper dispute form; chargeback / fraud / DD routes	PSR 2017 regs 76–77 [6]
Intl. beneficiary onboarding	Faxed IBAN and wet-ink verification; screening	MLR 2017 [8], sanctions/PEP
Balance certificate	Written request with postal wait; sealed certificate	Evidence document
Trusted devices & sessions	Branch visit to reset access; device/session control	SCA [3], SYSC [12]

SECTION 6

6 Regulatory control mapping

The workflows are designed around the controls a production implementation would require. The prototype surfaces each control in the user experience so that a reviewer can locate where an obligation is discharged. The mapping below is indicative of intent, not a statement of certified compliance.

Framework	Obligation reflected	Ref
Consumer Duty	Demonstrable good outcomes; fair treatment designed into journeys	[1]
PSD2 / SCA-RTS	Strong Customer Authentication and risk-based step-up before sensitive actions	[2][3]
Confirmation of Payee	Account-name verification before a first payment to a new payee	[4]
Direct Debit Guarantee	Immediate, guaranteed refund route for Direct Debit disputes	[5]
PSR 2017 (regs 76–77)	Refund of unauthorised transactions; correction of errors	[6]
APP reimbursement	Cancel window and reimbursement framing on outbound payments	[7]
MLR 2017 / FTR	Sanctions, PEP and adverse-media screening; payer/payee information	[8][9]
FCA DISP	Complaint intake, decision, and escalation to the Financial Ombudsman	[10]
FCA SYSC 9	Material actions recorded to an audit trail	[12]
HMRC MTD	Digital preparation and submission of VAT returns	[14]
UK GDPR	Data separation, ring-fencing and purpose limitation	[15]

SECTION 7

7 Engineering and quality approach

Although a concept artefact, the prototype is produced through a disciplined, repeatable pipeline. The pipeline exists because the failure modes of a large single-file front end are predictable, and because a credible artefact must actually work when demonstrated — not merely compile. Four stages are kept distinct:

- **Scaffold** — new workflows are generated from a template that encodes the file conventions and forecloses the recurring defects (state reset, dual-shell wiring, input handling).
- **Review** — a parallel, multi-lens review examines each change for architecture, security, engineering, design-system and state-integrity concerns before it is accepted.
- **Gate** — an automated pre-commit check blocks accessibility, colour-token, markup and build regressions from ever leaving the working tree.
- **Runtime verification** — the built application is driven headlessly through real journeys, and audited against WCAG 2.1 AA [16] on each run, so that behaviour is proven rather than assumed.

WHY THIS MATTERS

Static review confirms that code exists and is well-formed; only driving the running application confirms that a control actually fires. The pipeline is designed so that the artefact demonstrated to stakeholders is the artefact that has been verified.

Design system

A single visual language governs spacing, colour, typography and interaction: a fixed spacing scale, the Santander brand red reserved for the primary action on any screen, a warm neutral palette, tabular figures for every monetary value, and exactly one primary call-to-action per view. Consistency here is not decoration; it is what makes seventeen distinct journeys feel like one coherent product.

SECTION 8

8 Accessibility

Accessibility is treated as a build gate rather than a closing audit. Interactive controls carry visible focus states and accessible names; status is never signalled by colour alone; and the interface is measured against WCAG 2.1 AA [16] using an automated engine on every verification run. Where the automated audit identified shortfalls — colour-contrast on small text and unnamed icon-only controls — they were corrected and re-measured to zero on the core screens. Full conformance, including assistive-technology testing, is identified as a pre-production activity.

SECTION 9

9 Limitations and assumptions

The prototype should be read with its boundaries in mind, so that its evidence is neither over- nor under-weighted:

- It is front-end only: no data is stored, transmitted or processed, and no control is enforced server-side.
- All names, figures, accounts and references are illustrative and fictional.
- Regulatory behaviours are represented to communicate intent, not certified for production use.
- It has not been security-tested, load-tested, or assured to production standards.

SECTION 10

10 Indicative next steps

Should the proposition be endorsed, a proportionate path from concept to pilot would be:

- Prioritise the highest-value journeys — by volume, cost-to-serve and customer friction — for a backend-integrated pilot.
- Validate the shortlisted journeys with business customers through structured research, using the prototype as the discussion artefact.
- Complete a full WCAG 2.1 AA conformance review [16], including assistive-technology testing.
- Define the integration, security, resilience and data-protection requirements for a production build, and the target operating model to support it.

REFERENCES

References

[1] FCA, A New Consumer Duty — Policy Statement PS22/9 (July 2022); FCA Principle 12 and PRIN 2A.

[2] The Payment Services Regulations 2017 (SI 2017/752), transposing Directive (EU) 2015/2366 (PSD2).

- [3] Commission Delegated Regulation (EU) 2018/389 — Regulatory Technical Standards for Strong Customer Authentication and common and secure open standards of communication (SCA-RTS).
- [4] Pay.UK — Confirmation of Payee (CoP) scheme rules.
- [5] Bacs — the Direct Debit Guarantee.
- [6] The Payment Services Regulations 2017, regs 76–77 (unauthorised and incorrectly executed transactions).
- [7] PSR — Faster Payments APP scams mandatory reimbursement requirement (in force 7 October 2024).
- [8] The Money Laundering, Terrorist Financing and Transfer of Funds (Information on the Payer) Regulations 2017 (SI 2017/692).
- [9] Regulation (EU) 2015/847 on information accompanying transfers of funds (the Funds Transfer Regulation), as retained in UK law.
- [10] FCA Handbook — Dispute Resolution: Complaints (DISP); the Financial Ombudsman Service.
- [11] FCA Handbook — Banking: Conduct of Business Sourcebook (BCOBS).
- [12] FCA Handbook — Senior Management Arrangements, Systems and Controls (SYSC 9 record-keeping); the Senior Managers and Certification Regime (SM&CR;).
- [13] Consumer Credit Act 1974.
- [14] HMRC — Making Tax Digital for VAT.
- [15] UK GDPR and the Data Protection Act 2018 (as amended by the Data Use and Access Act 2025), Article 5(1)(c) — data minimisation and purpose limitation.
- [16] W3C — Web Content Accessibility Guidelines (WCAG) 2.1, Level AA.
- [17] ICO — guidance on the pre-employment vetting of candidates: necessity, proportionality and data minimisation when verifying individuals (relevant to the signatory ID register).

*Concept prototype — front-end only. No backend, no live system connections, and no real customer, account or payment data.
Prepared for internal discussion.*